

HU Collection

Test of Mathematics for University Admission

Mock Set J Paper 1

Applications of Mathematical Knowledge

Time allowed: 75 minutes

Questions: 20 multiple choice

Calculators: not permitted

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Name: _____

Date sat: _____

Score: _____ / 20

Instructions

Answer all 20 questions. Each question has exactly one correct option and is worth one mark. No marks are deducted for wrong answers. Every question uses AS content and the higher GCSE content that the TMUA assumes, including some of its less-practised corners: sectors and segments in radians, the ambiguous case of the sine rule, nets of cones, sums to infinity in context, and curves built from the modulus function. Several questions are longer than a typical TMUA question; the extra words are there to be read carefully.

[illegible][illegible]

Question 1

The cubic equation $x^3 + ax^2 + bx + c = 0$ has three real roots α , β and γ . The three numbers $\alpha + \beta$, $\beta + \gamma$ and $\gamma + \alpha$ are the roots of the equation

$$x^3 - 2x^2 - 5x + 6 = 0.$$

Find the value of $1 + a + b + c$.

- A -12
- B -7
- C -6
- D 0
- E 4
- F 6

Question 2

A chord AB of a circle with centre O and radius r divides the circle into two segments. The smaller of these, the *minor segment*, is bounded by the chord AB and the shorter arc AB . Its perimeter, the length of that arc plus the length of the chord, is exactly $\left(\frac{\pi}{3} + 1\right)r$. Find the ratio

$$\frac{\text{area of the minor segment}}{\text{area of triangle } OAB}.$$

- A $\frac{\pi - 3}{3}$
- B $1 - \frac{3\sqrt{3}}{2\pi}$
- C $\frac{4\sqrt{3}\pi - 9}{9}$
- D $\frac{9}{2\sqrt{3}\pi - 9}$
- E $\frac{2\sqrt{3}\pi - 9}{9}$
- F $\frac{2\pi - 3\sqrt{3}}{6}$

Question 3

A point P moves in the plane so that its distance from the point $(0, 2)$ is always exactly 1 greater than its distance from the x -axis. The path traced by P and the line $y = 2$ together enclose a finite region. Find its area.

- A 4
- B 6
- C 8
- D 9
- E 12
- F 3

Question 4

Amy and Ben play a game with a fair six-sided die. They take turns, starting with Amy, and on each turn the player rolls the die once. Amy wins as soon as she rolls a 6. Ben wins as soon as he rolls a 5 or a 6. There is one extra rule: a player who rolls a 1 immediately takes another turn, instead of passing the die to the other player. Find the probability that Amy wins.

- A $\frac{1}{3}$
- B $\frac{3}{8}$
- C $\frac{2}{5}$
- D $\frac{5}{12}$
- E $\frac{6}{13}$
- F $\frac{5}{13}$

Question 5

Find the complete set of real values of x for which

$$\log_{x^2}(x+2) < \frac{1}{2}.$$

- A $-2 < x < 1$, with $x \neq -1$ and $x \neq 0$
- B $0 < x < 1$
- C $-1 < x < 1$, with $x \neq 0$
- D $-2 < x < 1$, with $x \neq 0$
- E $-2 < x < -1$
- F $x < 1$, with $x \neq -1$ and $x \neq 0$

Question 6

The curve $y = x^4 - 4ax^3 + 6x^2$, where a is a constant, has a stationary point that is neither a local maximum nor a local minimum. Find the y -coordinate of this stationary point.

- A 0
- B 1
- C 2
- D 3
- E 9
- F 18

Question 7

Find the area of the region of the plane consisting of all the points (x, y) that satisfy

$$|2y - x^2 - 1| + |x| \leq 3.$$

- A $\frac{9}{2}$
- B 6
- C 9
- D 12
- E 18
- F $\frac{27}{2}$

Question 8

How many non-congruent triangles ABC are there in which $AB = 12$, angle $ACB = 45^\circ$, and the length BC is a whole number?

- A 12
- B 16
- C 17
- D 24
- E 28
- F 20

Question 9

A square has side length 2. A circle is drawn with the same centre as the square. Each side of the square crosses the circle twice, and the part of each side that lies inside the circle subtends an angle of 60° at the centre. Find the area of the region that lies inside both the square and the circle.

- A $\frac{4\pi}{9} + \sqrt{3}$
- B $\frac{4\pi}{9} + \frac{4\sqrt{3}}{3}$
- C $\frac{8\pi}{9} + \frac{2\sqrt{3}}{3}$
- D $\frac{4\pi}{3} - \frac{4\sqrt{3}}{3}$
- E $\frac{2\pi}{9} + \frac{4\sqrt{3}}{3}$
- F $\frac{8\pi}{9} + \frac{4\sqrt{3}}{3}$

Question 10

Find the total length of the set of values of x with $0 \leq x \leq 2\pi$ for which exactly two of the three numbers

$$\cos x, \quad \sin 2x, \quad \cos 3x$$

are positive.

- A $\frac{\pi}{3}$
- B $\frac{\pi}{2}$
- C $\frac{2\pi}{3}$
- D π
- E $\frac{5\pi}{6}$
- F $\frac{7\pi}{6}$

Question 11

A nature reserve plants 100 saplings in its first year, and in each later year it plants 5% more saplings than it planted the year before. At the end of every year, after that year's planting, one tenth of all the saplings then in the reserve die. Let P_n be the number of saplings planted in year n , and let T_n be the number of living saplings at the end of year n , after that year's deaths. Treat all these quantities as exact real numbers, ignoring rounding. As n becomes large, the ratio T_n/P_n approaches a limit. Find this limit.

- A $\frac{63}{10}$
- B 6
- C 7
- D $\frac{15}{2}$
- E 9
- F $\frac{21}{2}$

Question 12

The polynomial p has degree 3, and

$$\int_0^1 p(x) dx = \int_0^2 p(x) dx = \int_0^3 p(x) dx = 0, \quad p(0) = 6.$$

Find $p(4)$.

- A -26
- B -24
- C 24
- D -50
- E 50
- F 26

Question 13

A circular sheet of paper of radius 3 is cut along two radii into two sectors. Each sector is then rolled up, without overlapping, to form the curved surface of a cone, so that the two straight edges of the sector are joined and become a slant height of the cone. The height of one cone turns out to be exactly twice the height of the other. Find

$$\frac{\text{volume of the taller cone}}{\text{volume of the shorter cone}}.$$

- A 2
- B $\frac{11 - 3\sqrt{13}}{4}$
- C $\frac{\sqrt{13} - 3}{4}$
- D $\frac{11 - 3\sqrt{13}}{8}$
- E $11 + 3\sqrt{13}$
- F $\frac{1}{2}$

Question 14

A farmer has exactly 12 metres of fencing, which she uses to enclose a pen in the shape of a triangle. One of the angles of the triangle must be 120° . Find the greatest possible area of the pen, in square metres.

- A $4\sqrt{3}$
- B $36(7 - 4\sqrt{3})$
- C $252\sqrt{3} - 432$
- D $126\sqrt{3} - 216$
- E $3\sqrt{3}$
- F $144(2 - \sqrt{3})^2$

Question 15

How many real solutions does the equation

$$2|\sin x| = \left| \frac{|x|}{\pi} - 2 \right|$$

have? (Here x is measured in radians.)

- A** 7
- B** 12
- C** 13
- D** 15
- E** 16
- F** 14

Question 16

A rectangle is drawn with one side on the x -axis and its other two vertices on the curve

$$y = 3 - |x^2 - 1|,$$

in such a way that the whole rectangle lies in the region enclosed by the curve and the x -axis. Find the greatest possible area of such a rectangle.

- A** 6
- B** $\frac{32\sqrt{3}}{9}$
- C** $\frac{16}{3}$
- D** $2\sqrt{6}$
- E** $4\sqrt{2}$
- F** $3\sqrt{3}$

Question 17

The function F is defined for all real x by

$$F(x) = \int_0^x (t-1)|t-3| dt.$$

The graph of $y = F(x)$ and the x -axis enclose exactly one finite region. Find its area.

- A $\frac{9}{4}$
- B $\frac{4}{3}$
- C $\frac{3}{4}$
- D $\frac{27}{4}$
- E $\frac{9}{2}$
- F 3

Question 18

Find the area of the region of the plane consisting of all the points (x, y) that satisfy both

$$|x| + |y| \leq 3 \quad \text{and} \quad y \geq |x^2 - 1|.$$

- A $\frac{11}{3}$
- B $\frac{17\sqrt{17} - 63}{6}$
- C $\frac{17\sqrt{17} - 41}{12}$
- D $\frac{17\sqrt{17} - 41}{6}$
- E $\frac{17\sqrt{17} - 25}{6}$
- F 5

Question 19

Find the area of the region of the plane that lies inside the circle $x^2 + y^2 = 4$ and above the parabola $y = x^2 - 2$.

- A $\frac{4\pi}{3} + \sqrt{3}$
- B $\frac{8\pi}{3} + \sqrt{3}$
- C $\frac{4\pi}{3} + 3\sqrt{3}$
- D $2\pi + 2\sqrt{3}$
- E $\frac{8\pi}{3} - 3\sqrt{3}$
- F $\frac{4\pi}{3} + 2\sqrt{3}$

Question 20

For a real constant k , consider the equation

$$|x^2 - 6|x| + 5| = x + k.$$

Find the sum of all the values of k for which this equation has exactly three real solutions.

- A $\frac{29}{4}$
- B $\frac{25}{4}$
- C $\frac{17}{2}$
- D $\frac{15}{2}$
- E $\frac{45}{4}$
- F $\frac{5}{4}$